

Altera Cyclone III Programming Procedure

3 GEC-SE

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Altera Cyclone III Programming Procedure

1.0 Introduction

This document contains instructions on how to program the Altera Cyclone III FPGA. This will be done using the Altera Cyclone III starter board and a computer with the most recent version of the Quartus II software.

2.0 Parts list

1. Altera Cyclone III Starter Board (See Figure 1)
2. 12V DC Power Supply
3. USB Cable -Type A Male to Type B Male

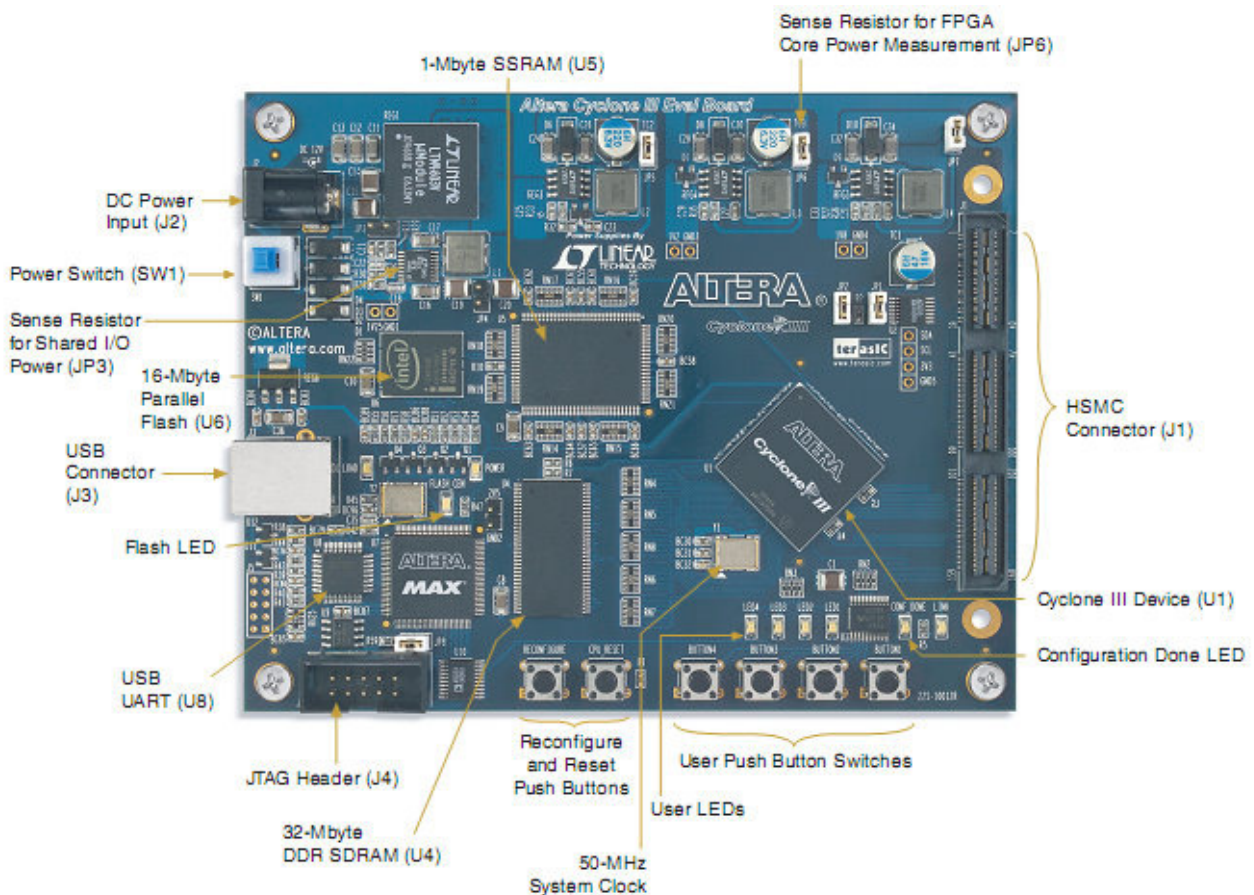


Figure 1 : Altera Cyclone III Starter Board

3.0 Setting Up the Project in Quartus II

Perform the following steps to set up the project.

1. Open the Quartus II Software.
2. When the software opens, click on the “Create a New Project” Button as shown in Figure 2 below.

Note: If you already have a project created, click “Open then Existing Project”, select the desired project file and skip to Section 4.0.



Figure 2: Click on the Create a New Project tab.

3. The “Introduction” window will appear. Click “Next>”.
4. The “Directory, Name, Top Level Entity” window will appear. Enter the Directory, name of the Project and the name of the Top Level Entity as shown in Figure 3 below. Make sure that the Top Level Entity name matches the exact name of the design file. Click “Next>”

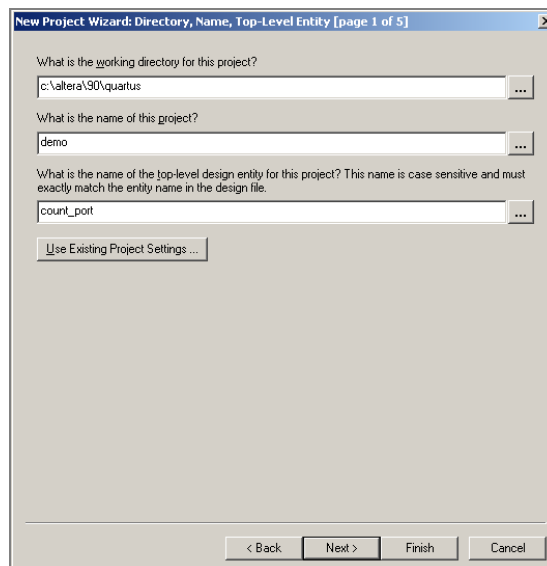


Figure 3: Enter the directory, name of the project and top level entity

5. The “Add Files” window will appear. Enter the design file to be programmed. If there is no design file created, leave this section blank. Click “Next>”

6. The “Family and Device Settings” will appear. Enter the settings for the Family, Package, Pin Count and Speed Grade as shown in Figure 4 below. For the Altera Cyclone III Started board, enter the following settings and click “Next>”:
 - a. Family > Cyclone III
 - b. Package > FBGA
 - c. Pin Count > 324
 - d. Speed Grade > Fastest
 - e. Name > EP3C25F324C6

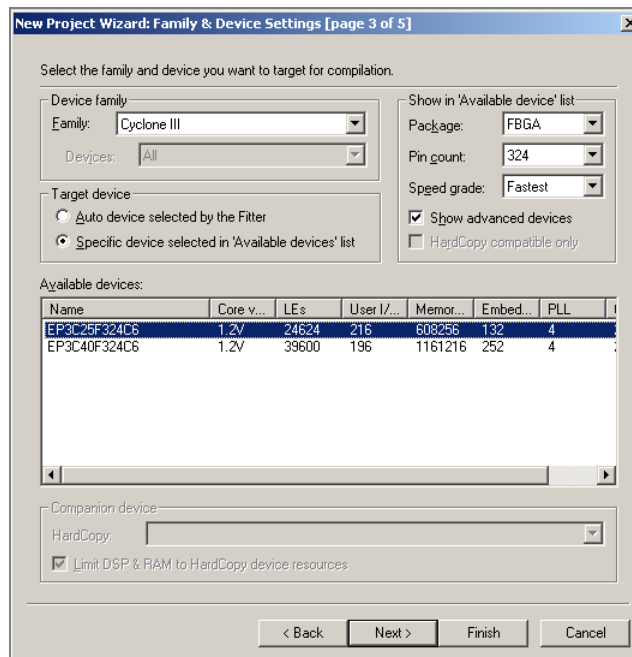


Figure 4: Enter the settings for the family, package, pin count and speed grade

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7. The “EDA Tool Settings” window will appear. Select the desired tools and click “Next>”.
8. The “Summary” window will appear. Verify that the settings are correct. Click “Finish”.

4.0 Pin Assignments

Perform the following steps to properly assign the inputs and outputs to pins on the device:

1. In the “Task” window, double click on “Analysis and Synthesis” as shown in Figure 5 below.

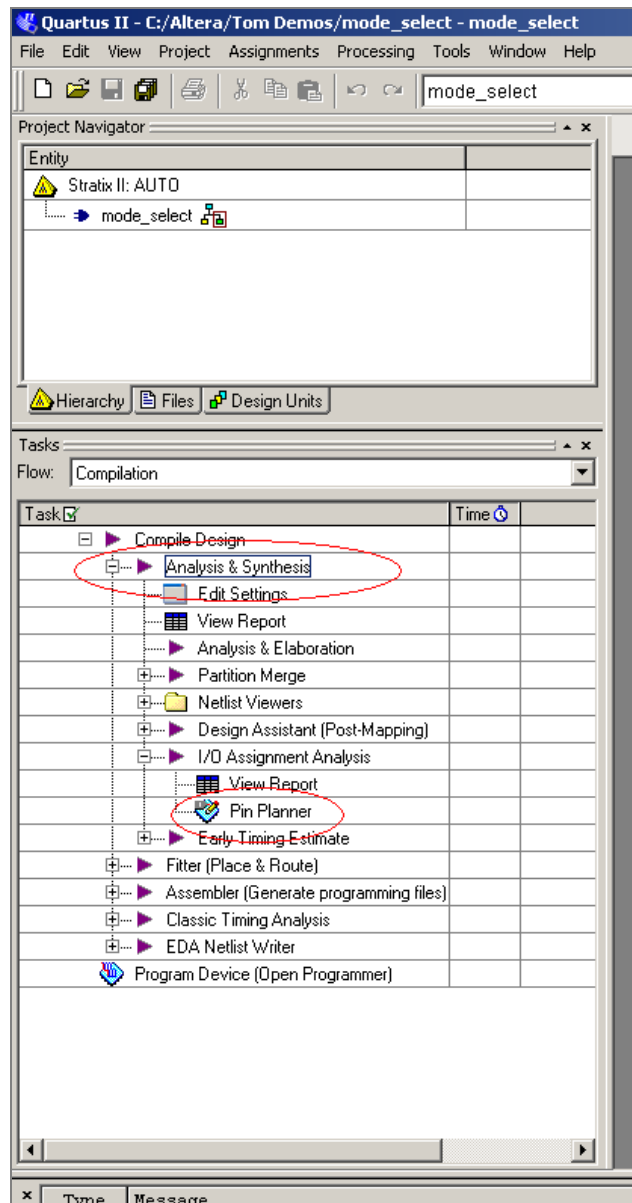


Figure 5: Double Click on "Analysis and Synthesis" and then Double click on "Pin Planner"

2. When the compilation is finished, double click on “Pin Planner” which is also located in the “Task” window as shown in Figure 5.

- The “Pin Planner” window should appear. Assign the pins to the proper location using the pull down menu located under the “location tab” as shown in Figure 6 below.

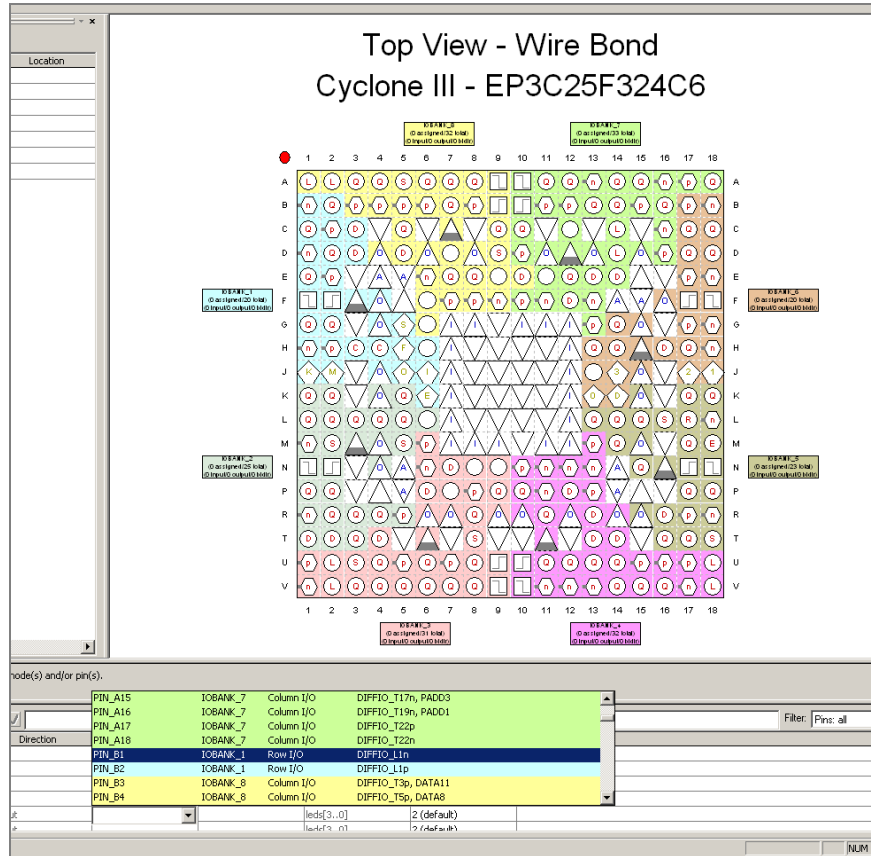


Figure 6: Pin Planner and Pin selection drop down menu

- Close this window when all pins have been assigned.

5.0 Programming the FPGA

Perform the following steps to program the FPGA:

1. Turn on the Altera Cyclone III Starter Board and connect the USB cable from the board to the PC.
2. Compile the design by clicking on “Compile Design”. This is located in the “task” window.
3. Once the compilation is complete, click on “Program Device (open programmer)” located in the “task” window.
4. The window shown in Figure 7 should appear. Click on the “Hardware Setup” button.

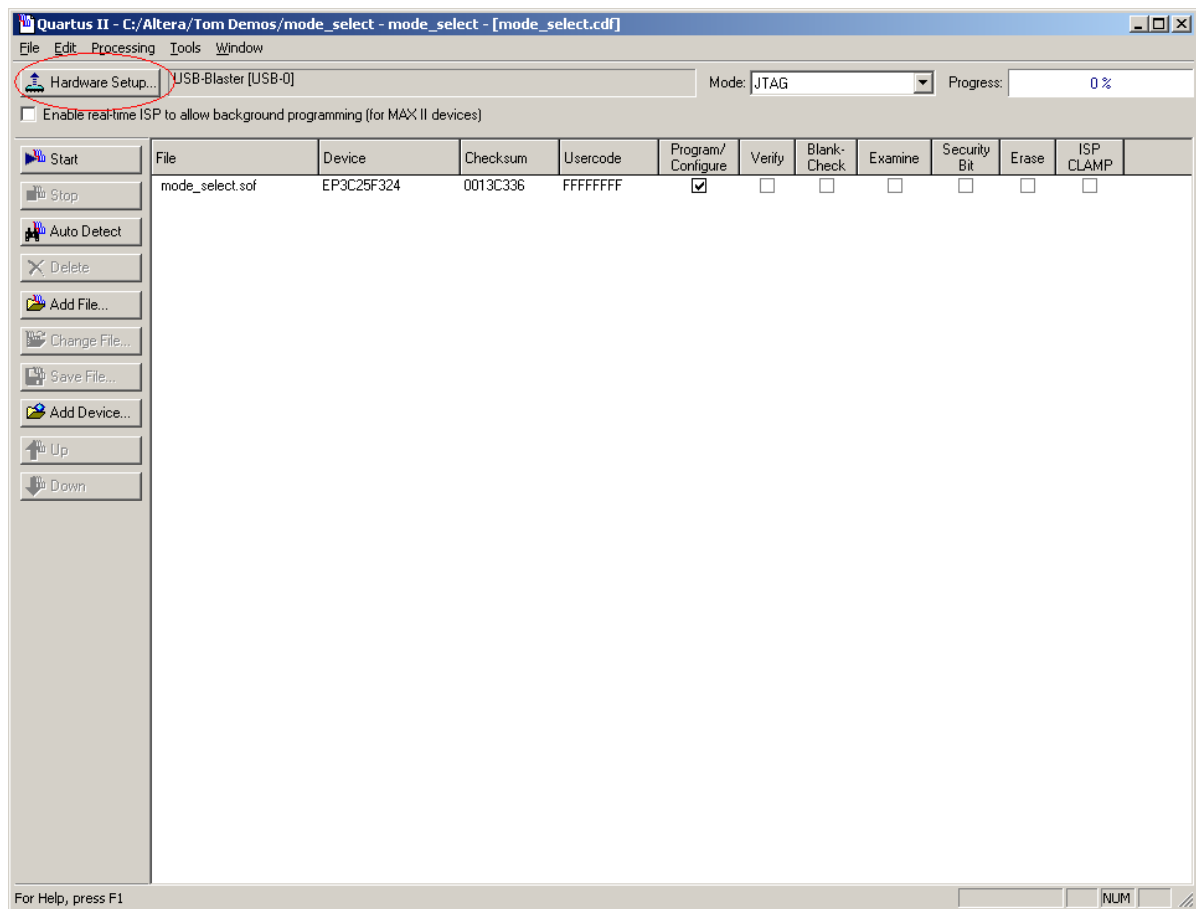


Figure 7 : Click on the "hardware setup" button

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5. The “Hardware Setup” window should appear as shown in Figure 8. Select “USB-blaster (USB-0)” and click “Close”.

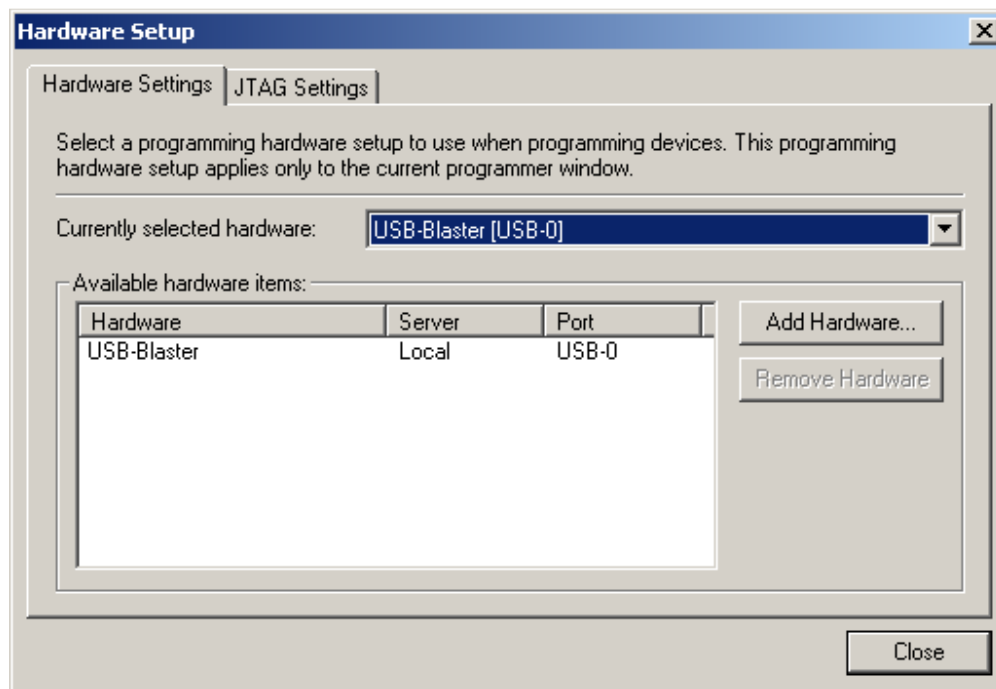


Figure 8 : The “Hardware Setup” window

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6. Select the mode from the pull down menu in the programming window.
 7. Highlight the .sof file and click “Start”.

Développement d'un circuit sur FPGA

Elaboration du matériel:

On se contentera ici de suivre scrupuleusement le texte du **My First FPGA Design** (*tt_my_first_fpga.pdf*) avec cependant les nombreux aménagements suivants :

- ✓ La carte cible est Cyclone III Starter Board d'Altera (EP3C25F324C8).
- ✓ Page 10 : créer un fichier VHDL au lieu d'un fichier Verilog par le choix **File >New > VHDL File** puis enregistrer le fichier sous le nom **simple_counter.vhd**
- ✓ Page 11 : copier le code VHDL ci-dessous dans le fichier **simple_counter.vhd** au lieu du code donné en verilog dans la page 11.

```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_unsigned.all;
entity counter is
port (clock: in std_logic;
counter_out: out std_logic_vector(31 downto 0));
end counter;
architecture A_counter of counter is
signal counter: std_logic_vector(31 downto 0):=(others=>'0');
begin
count:process(clock)
begin
if (clock'event and clock='1') then
counter<=counter+'1';
end if;
end process;
counter_out<=counter;
end A_counter;
```

Travail demandé :

Réaliser toutes les étapes de conception décrites dans le tutorial

(*tt_my_first_fpga.pdf*)

Après compilation et simulation fonctionnelle, déduire le rôle et le fonctionnement de ce circuit.